University of Santo Tomas – Faculty of Engineering
Electronics Engineering Department
ECE 2112: Advanced Computer Programming and Algorithms

EXPERIMENT 5: INTRODUCTION TO MATLAB PROGRAMMING

Name: Surname, First Name M.I. **Date Submitted:** _____

Section: 2ECE-_____

I. Intended Learning Outcomes

At the end of this laboratory activity, the student should be able to:

1. create real and complex matrices in MATLAB;
2. distinguish transpose from complex-conjugate transpose operations;
3. express a simultaneous linear system in matrix form; and
4. solve and check a linear system using the inverse method and Cramer's Rule.

II. Instructions

Write a MATLAB script containing clearly labeled sections for both problems. Display the requested matrices and numerical checks. Use MATLAB matrix operations and built-in functions; do not type a computed answer directly into the output.

III. Programming Problems

A. TRANSPOSE OPERATIONS

Define the following matrices exactly as shown:

$$R = \begin{bmatrix} 5 & -2 & 0 & 7 \\ 1 & 3 & -4 & 2 \\ 6 & 0 & 8 & -1 \end{bmatrix}, \qquad Z = \begin{bmatrix} 2+3i & -1+4i & 5-2i \\ -3+i & 6 & 2+5i \end{bmatrix}.$$

For each matrix, compute and display all three expressions:

```
transpose(A)
A'
A.'
```

where `A` is replaced by `R` and then by `Z`.

Answer the following in MATLAB comments below your output:

a. Which operations give identical results for the real matrix `R`?
b. Which two operations give identical results for the complex matrix `Z`?
c. What additional operation is performed by the apostrophe operator `'` when the matrix contains complex entries?

B. TWO METHODS FOR A LINEAR SYSTEM

Solve the simultaneous system

$$\begin{aligned} 2x_1 - x_2 + 3x_3 + 2x_4 &= 16, \\ x_1 + 4x_2 - 2x_3 + x_4 &= -7, \\ 3x_1 + 2x_2 + x_3 - x_4 &= 6, \\ -2x_1 + 3x_2 + 4x_3 + 2x_4 &= 7. \end{aligned}$$

Create the coefficient matrix `A` and right-hand-side vector `b`. Then compute the solution in two ways.

1. **Inverse method:** compute `x_inv` using the inverse of `A`.
2. **Cramer's Rule:** replace one column of `A` at a time with `b`, use determinants, and store the result in `x_cramer`. A loop is optional.

Do not use MATLAB's left-division operator as either of the two required methods.

Required checks: Display both solution vectors and `det(A)`. Also display `norm(x_inv-x_cramer)` and `norm(A*x_inv-b)`. The determinant must be nonzero. Both norm checks should be approximately zero to numerical precision.

IV. Submission Requirements

Submit one MATLAB script named `Surname_PA5.m`. Include your name and section in comments, clearly labeled Problem A and Problem B sections, all required displays, and the three short comments explaining the transpose behavior. The script must run from beginning to end without errors.

V. Grading

Component	**Points**
Transpose Operations	35
Inverse Method	25
Cramer's Rule and Numerical Checks	30
Code readability and organization	10
Total	**100**

End of Experiment 5